CA-OPD: Conflict-Averse Multi-Teacher On-Policy Distillation for Diffusion Models

Anonymous Author(s)
Affiliation email

Abstract

On-policy distillation (OPD) has recently emerged as an efficient paradigm for multi-task alignment of diffusion models: task-specific teachers supervise a single student along its own rollout trajectory through a closed-form, low-variance per-step velocity-matching loss. In the multi-teacher setting, DiffusionOPD sums the per-teacher losses uniformly, which drives the student toward the per-step *mean* of the teacher velocity fields. We revisit this choice through the lens of multi-objective optimization. Our first contribution is a rigorous theory: the multi-teacher OPD losses are values of a *single* functional—squared velocity error in a shared isotropic output metric, evaluated at shared on-policy states—so they are commensurable across teachers, and adaptive weighting is well-posed in a way it is not for generic multi-task learning. We show that the uniform mean is Pareto-optimal in *loss* space (so the naive convex-hull argument does not apply), and locate the real difficulty in the nonlinear loss→reward map, under which the loss-sum minimizer can be reward-dominated. We then propose **CA-OPD**, a lightweight conflict-averse reweighting that replaces the uniform sum with an entropy-regularized (temperature-τ) Tchebycheff min-max over EMA teacher losses; an envelope-theorem argument shows the detached-weight gradient is *exact* for the smooth surrogate, and $\tau \to \infty$ recovers uniform OPD. Empirically, on a faithful reproduction of DiffusionOPD (SD3.5-Medium, three teachers) whose numbers match the original, CA-OPD improves *convergence efficiency* on the lagging (aesthetics) axis—reaching the uniform baseline's best aesthetic score roughly 180 steps earlier—and exposes τ as a knob trading convergence speed against final quality. Honestly, on this near-saturated three-teacher benchmark CA-OPD does *not* improve final quality—at convergence it is within single-seed noise of the uniform baseline on every reward axis and higher on none. We further ablate four principled ideas (min-max, Huber-robust, Min-SNR weighting, SDE→ODE annealing), each with a mechanistic explanation, mapping the design space of OPD improvements.

1 Introduction

Reinforcement learning has become a standard tool for aligning diffusion-based text-to-image models to reward signals such as compositional correctness, text rendering, and aesthetic quality [12, 16, 23]. Most methods optimize a *single* reward, yet users expect one model to satisfy many objectives at once. Directly training on a mixture of rewards suffers from cross-task interference and difficulty imbalance, while sequential (cascade) RL is cumbersome and prone to catastrophic forgetting.

On-policy distillation (OPD) side-steps both problems. DiffusionOPD [10] first trains one specialized teacher per task, then distills all teachers into a single student along the student's *own* denoising rollout. Its key technical observation is that, at each reverse step, student and teacher

one-step Gaussian transition kernels are evaluated at the same on-policy state and share the same covariance, so their reverse KL is closed-form and reduces to a dense, low-variance velocity-matching loss. With M teachers, DiffusionOPD simply *sums the per-teacher losses uniformly*.

This paper. We ask what the uniform sum is actually optimizing, and whether a principled reweighting can do better. Uniform summation is *linear scalarization* of the per-teacher objectives; a textbook concern is that linear scalarization only reaches the convex hull of a Pareto front and is dominated on non-convex regions. We show—and an independent verifier confirmed—that this concern is *misplaced in loss space*: under OPD's shared isotropic output metric the per-teacher losses are convex quadratics, the achievable loss set is convex, and the uniform (teacher-mean) student is genuinely Pareto-optimal in loss. The real difficulty is elsewhere: the mapping from velocity-matching loss to task *reward* is nonlinear and non-monotone (rewards are read off integrated samples, reward models saturate and can be hacked, and a finite shared LoRA couples tasks), so the loss-sum minimizer—though loss-optimal—can be *reward-dominated*. This recast (from "non-convex loss hull" to "reward-space frontier selection") is the honest and correct motivation for moving off the uniform point.

Building on this, we make four contributions.

- **A rigorous multi-objective theory for multi-teacher OPD** (§4). We formalize multi-teacher OPD as multi-objective optimization, prove the teacher-mean is loss-Pareto-optimal, relocate the difficulty to the loss→reward map, and introduce an adaptive Tchebycheff (min-max) scalarization. We prove via the envelope (Danskin) theorem that the *detached*-weight gradient used in practice is *exact* for the smooth log-sum-exp surrogate, and that the per-teacher losses are commensurable because they are one functional in a shared output metric—the property that makes adaptive weighting well-posed here but not in generic multi-task learning. This theory is the strongest part of the paper.
- **CA-OPD** (§3), a lightweight conflict-averse reweighting: weight each teacher's loss by a softmax over an EMA of its per-step KL, up-weighting the currently worst-matched teacher. A temperature τ interpolates from pure min-max ($\tau \to 0$) to uniform DiffusionOPD ($\tau \to \infty$), so mopd lies in CA-OPD's search space. CA-OPD improves *convergence efficiency* on the lagging reward axis and exposes τ as a speed-vs-ceiling knob.
- **A faithful reproduction** of DiffusionOPD on SD3.5-Medium with three teachers (§5.2). Our base, teacher, and distilled-student numbers match the original within evaluation noise, and our student cross-checks against the released official student—a triple-validated reproduction that is itself a useful contribution.
- **An honest ablation** of four principled OPD-improvement ideas (§6): min-max (faster early), Huber-robust (neutral), Min-SNR weighting (hurts), and SDE→ODE annealing (hurts most), each with a mechanistic explanation. This maps the design space of OPD modifications and shows CA-OPD's early-convergence benefit is non-trivial.

What we do *not* claim. We are explicit that on this near-saturated three-teacher benchmark CA-OPD does *not* beat uniform DiffusionOPD on final quality: at convergence the two are within single-seed noise on every reward axis, and CA-OPD is higher on none. CA-OPD's demonstrated value is convergence efficiency on the hardest axis and per-axis controllability through τ, not a new state of the art. We discuss when stronger gains should be expected—higher-conflict teacher sets, or regret-based weights $r_m = \mathcal{L}_m - \text{floor}_m$—as future work (§8).

2 Related Work

RL and distillation for diffusion alignment. Reinforcement learning has been widely adapted to align diffusion text-to-image models to reward signals [12, 16, 23]. Flow-GRPO [12] casts flow-matching sampling as an MDP and optimizes a GRPO-style objective; DiffusionNFT [23] performs online reinforcement on the forward process. These methods excel at single rewards but struggle to serve several objectives jointly. DiffusionOPD [10] reframes multi-task alignment as *on-policy distillation*: specialized teachers, trained once per task, supervise a single student along its own rollout via a closed-form per-step KL, extending the OPD idea [18] from tokens to continuous diffusion states. We take DiffusionOPD as our base method, reproduce it faithfully, and study how the multiple teachers should be combined.

Multi-objective and multi-task optimization. The limitations of uniform loss summation are classical. Linear scalarization recovers only the convex hull of the Pareto front [1, 13], whereas weighted Tchebycheff (min-max) scalarization can reach the entire front, including non-convex regions [13]. In deep multi-task learning, MGDA [2, 17] finds a common descent direction with a Pareto-stationarity guarantee, PCGrad [22] projects away conflicting gradient components, and CA-Grad [11] takes a conflict-averse step that reduces the worst-case local objective while converging to a stationary point of the average loss. These methods generally assume commensurable objectives, an assumption violated by generic MTL where losses have incompatible units and scales. A central theoretical point of our work (§4) is that multi-teacher OPD *does* satisfy commensurability—all per-teacher losses are the same velocity-error functional in one shared isotropic output metric—so adaptive Tchebycheff scalarization is particularly well-posed here. We adopt scalarization rather than gradient surgery and are careful not to over-transfer output-space isotropy to parameter space.

Loss weighting in diffusion training. Timestep-dependent loss weighting is common in diffusion training; Min-SNR [5] treats denoising as multi-task learning across noise levels and down-weights low-noise (high-SNR) steps to balance the objective. We test an analogous per-step reweighting inside OPD (§6) and find it harmful, because high-noise steps set global structure that OPD must preserve—a useful negative result that clarifies why OPD's uniform per-step weighting is a good default. Robust regression via the Huber loss [7] motivates our trust-region variant on the velocity residual, which we find neutral.

Evaluation. We evaluate with the same reward suite as the base method: GenEval [4] and OCR (rule-based), and PickScore [8], ClipScore [6], HPSv2.1 [20], Aesthetics [15], ImageReward [21], and UnifiedReward [19] (model-based), on DrawBench [14] prompts, with SD3.5-Medium [3] as the base model.

3 Method

We first recap the base DiffusionOPD objective (§3.1), then present CA-OPD, our conflict-averse multi-teacher reweighting (§3.2). The full multi-objective theory—including the Pareto/reward-space analysis, the envelope-theorem exactness result, and the commensurability property—is developed in §4, which this section inputs directly.

3.1 Recap: DiffusionOPD

We adopt DiffusionOPD [10] verbatim as our base method. A flow-matching student v_θ rolls out its own reverse-time trajectory $x_{t_0}, \dots, x_{t_N}$ ($1{=}t_0{>}\cdots{>}t_N{=}0$) via the Euler–Maruyama discretization of Flow-GRPO [12]. At each step, the student and a frozen teacher v_ϕ define one-step Gaussian transition kernels that (i) are evaluated at the *same* on-policy state x_{t_j} and (ii) share the *same* covariance $\bar{\sigma}_j^2 I_d$ (a function only of the scheduler and global noise level). They therefore differ only in mean, and the per-step reverse KL is closed-form,

$$\mathrm{KL}\big(\mathcal{N}(\mu_S, \bar{\sigma}_j^2 I) \,\|\, \mathcal{N}(\mu_T, \bar{\sigma}_j^2 I)\big) = \frac{\|\mu_S(x_{t_j}) - \mu_T(x_{t_j})\|_2^2}{2\bar{\sigma}_j^2} \;\propto\; \|v_\theta - v_\phi\|_2^2, \tag{1}$$

introducing no Monte-Carlo variance. This yields dense, pathwise per-step velocity matching; in the deterministic ODE regime (the default) it becomes the direct L_2 loss $\frac{1}{2}\|\mu_S - \mu_T\|_2^2$ [10].

Uniform multi-teacher OPD (mopd). Given M frozen task teachers $\{v_{\phi_m}\}_{m=1}^M$, DiffusionOPD defines the per-teacher loss $\mathcal{L}_m(\theta)$ from (1) and trains on their *uniform* average, implemented as round-robin gradient accumulation with factor $G = M$:

$$\mathcal{L}_{\mathrm{mopd}}(\theta) = \frac{1}{M}\sum_{m=1}^{M} \mathcal{L}_m(\theta). \tag{2}$$

As we show in §4, the minimizer of (2) drives the student velocity toward the per-step teacher mean $\frac{1}{M}\sum_m v_{\phi_m}$.

3.2 CA-OPD: conflict-averse multi-teacher reweighting

CA-OPD replaces the uniform average (2) by an entropy-regularized Tchebycheff (min-max) scalarization over per-teacher reference-normalized losses. Because per-batch losses are noisy, we drive the weights from an exponential moving average of each teacher's per-step KL,

$$\bar{\ell}_m^{(t)} = \beta\, \bar{\ell}_m^{(t-1)} + (1-\beta)\, \mathcal{L}_m(\theta^{(t)}), \qquad w_m^{(t)} = \frac{\exp\big(\bar{\ell}_m^{(t)}/(\tau r_m)\big)}{\sum_k \exp\big(\bar{\ell}_k^{(t)}/(\tau r_k)\big)}, \tag{3}$$

and take the *detached*-weight loss

$$\mathcal{L}_{\text{CA-OPD}}^{(t)}(\theta) = \sum_{m=1}^{M} \mathrm{sg}\big[w_m^{(t)}\big]\, \mathcal{L}_m(\theta), \tag{4}$$

where $\mathrm{sg}[\cdot]$ is stop-gradient. This up-weights the currently *worst-matched* (largest EMA-KL) teacher. Two properties, proved in §4, make this principled rather than heuristic: (i) by the envelope (Danskin) theorem, (4) is the *exact* gradient of the smooth log-sum-exp Tchebycheff surrogate $\mathcal{F}_\tau(\theta) = \tau \log \sum_m \exp(\mathcal{L}_m/(\tau r_m))$ (no term through the softmax is missed); and (ii) as $\tau \to \infty$, $w_m \to 1/M$ and CA-OPD reduces *exactly* to mopd (2), so uniform DiffusionOPD is the $\tau \to \infty$ special case and lies inside CA-OPD's search space. The temperature τ controls how aggressively the worst teacher is prioritized.

Algorithm 1: CA-OPD (Stage 2 multi-teacher on-policy distillation)

Input: Teachers $\{v_{\phi_m}\}_{m=1}^{M}$; student $v_\theta \leftarrow v^{\text{ref}}$; temperature τ; EMA β; refs $\{r_m\}$.
Initialize EMA $\bar{\ell}_m \leftarrow 0$;
for *each training round* **do**
  for $m = 1, \dots, M$ **do**
    Sample prompts $\mathbf{c} \sim \mathcal{C}^{(m)}$; roll out student to get $\{x_{t_j}\}$; `// no_grad`
    Compute $\mathcal{L}_m(\theta)$ from Eq. (1) (ODE L_2 by default);
    Update EMA: $\bar{\ell}_m \leftarrow \beta \bar{\ell}_m + (1-\beta)\,\text{detach}(\mathcal{L}_m)$;
  end
  Weights $w_m \leftarrow \text{softmax}_m(\bar{\ell}_m/(\tau r_m))$; `// detached`
  Backprop $\sum_m \text{sg}[w_m]\,\mathcal{L}_m$ and take one optimizer step;
end

Implementation. CA-OPD is a small addition to the DiffusionOPD trainer and changes nothing about teacher training or the rollout. In each round-robin cycle we tag each accumulated micro-batch with its `teacher_idx`, maintain the per-teacher EMA `teacher_kl_ema`, and scale each teacher's contribution by the detached softmax weight (3) before the single backward pass. We use τ=0.5 and β=0.9; the reference r_m is fixed. Setting the flag off recovers the unmodified mopd baseline exactly (`config.train.ca_opd`, with `ca_opd_tau` and `ca_opd_ema` defaulting via `getattr`), so the two arms differ only in the weighting of an otherwise identical loss. Algorithm 1 summarizes Stage 2; Stage 1 (per-task teacher training) is unchanged from DiffusionOPD.

4 Theory: Multi-Teacher OPD as Multi-Objective Optimization

We recap the per-step closed-form supervision of DiffusionOPD [9] and its uniform multi-teacher extension (§4.1), formalize multi-teacher OPD as a multi-objective problem (§4.2), and then make our central point: in *loss* space the uniform (teacher-mean) student is already Pareto-optimal, so the difficulty is *not* a convex-hull artifact of loss scalarization; the difficulty lives in the nonlinear, non-monotone map from velocity-matching loss to task *reward*, under which the loss-sum minimizer can be reward-dominated (§4.3). We then introduce our adaptive Tchebycheff / min-max fix and prove its detached-weight gradient is exact via the envelope theorem (§4.4), give the output-space commensurability property that makes min-max scalarization meaningful in OPD (§4.5), discuss convergence and weighting risks honestly (§4.6), and close with a rigor remark on why symmetrizing the OPD divergence is vacuous (§4.7).

4.1 Recap: per-step closed-form KL and uniform multi-teacher OPD

We adopt the DiffusionOPD construction verbatim [9]. A flow-matching student v_θ rolls out its *own* reverse-time trajectory $x_{t_0}, \dots, x_{t_N}$ ($1{=}t_0{>}\cdots{>}t_N{=}0$) via the Euler–Maruyama discretization of Flow-GRPO [12]. At each step the student and teacher one-step transition kernels are Gaussians that (i) are evaluated at the *same* on-policy state x_{t_j} and (ii) share the *same* covariance $\bar{\sigma}_j^2 I_d$, which depends only on the scheduler $(t_j, \Delta t_j)$ and the global noise level. Hence they differ only in their means, and the per-step reverse KL between two equal-covariance Gaussians is closed-form,

$$\text{KL}\big(\mathcal{N}(\mu_S, \bar{\sigma}_j^2 I) \,\|\, \mathcal{N}(\mu_T, \bar{\sigma}_j^2 I)\big) = \frac{\|\mu_S(x_{t_j}) - \mu_T(x_{t_j})\|_2^2}{2\bar{\sigma}_j^2}, \tag{5}$$

introducing no Monte-Carlo variance [9]. Because student and teacher means share the identical affine map from velocity to transition mean, their difference is proportional to the velocity residual,

$$\mu_S(x_{t_j}) - \mu_T(x_{t_j}) = c_j \left(v_\theta(x_{t_j}, t_j) - v_\phi(x_{t_j}, t_j)\right), \qquad c_j := \left(1 + \frac{\sigma_{t_j}^2 (1-t_j)}{2t_j}\right)\Delta t_j, \tag{6}$$

so the per-step objective reduces to weighted velocity matching. Writing $w_j := c_j^2/2 > 0$ for the ODE regime (resp. $c_j^2/(2\bar{\sigma}_j^2)$ for SDE), the single-teacher OPD loss is

$$\mathcal{L}^\phi(\theta) = \mathbb{E}_{x_{0:N} \sim p_{S,\theta}} \left[\sum_{j=0}^{N-1} w_j \left\| v_\theta(x_{t_j}, t_j) - v_\phi(x_{t_j}, t_j) \right\|_2^2 \right] \;\propto\; \|v_\theta - v_\phi\|_2^2. \tag{7}$$

Multi-teacher (uniform round-robin). Given M frozen task teachers $\{v_{\phi_m}\}_{m=1}^M$, DiffusionOPD defines the per-teacher loss $\mathcal{L}_m(\theta) := \mathcal{L}^{\phi_m}(\theta)$ and trains on their uniform average via gradient accumulation $G = M$: $\mathcal{L}_{\text{total}}(\theta) = \frac{1}{M}\sum_{m=1}^M \mathcal{L}_m(\theta)$. Throughout we make the standing assumption implicit in DiffusionOPD, whose rollouts are generated under `no_grad`:

Assumption 1 (On-policy state distribution frozen within an update)**.** Within a single parameter update, the trajectory-sampling distribution $p_{S,\theta}$ is treated as a fixed measure p_S (the current student's rollout law); gradients flow only through the per-step velocity heads $v_\theta(x_{t_j}, t_j)$, not through the law of $x_{0:N}$.

4.2 Two objectives: matching loss vs. task reward

Two vector-valued quantities coexist, and conflating them is the source of the multi-teacher difficulty.

Definition 1 (Loss and reward objectives)**.** The *matching-loss* vector is $\mathcal{L}(\theta) = (\mathcal{L}_1(\theta), \ldots, \mathcal{L}_M(\theta)) \in \mathbb{R}_{\geq 0}^M$ with achievable region $\mathcal{A}_L = \{\mathcal{L}(\theta) : \theta \in \Theta\}$. The *task-reward* vector is $R(\theta) = (R_1(\theta), \ldots, R_M(\theta)) \in \mathbb{R}^M$, where R_m is the (evaluation) reward on task m; we want to *maximize* every R_m. Crucially, R is what we ultimately care about, whereas $\mathcal{L}$ is the optimized proxy.

Definition 2 (Pareto notions)**.** For a minimization objective, θ *dominates* θ' if it is componentwise no worse and strictly better in some component; a point is *Pareto-optimal* if dominated by none. A point is *Pareto-stationary* [2, 17] if $\sum_m \alpha_m \nabla \mathcal{L}_m = 0$ for some $\alpha \in \Delta^{M-1}$ (the simplex) — the first-order (KKT) necessary condition. The same notions apply to reward-maximization with inequalities reversed.

4.3 The uniform sum is loss-optimal but reward-suboptimal

We first record that, at the level of the *proxy loss*, uniform OPD has no defect.

Proposition 1 (Uniform OPD ⇒ teacher-mean velocity, and it is loss-Pareto-optimal)**.** *Under Assumption 1, in the pointwise (sufficient-capacity) regime the unique minimizer of* $\frac{1}{M}\sum_m \mathcal{L}_m$ *sets the student velocity at every* (x_{t_j}, t_j) *to the arithmetic mean* $v_\theta^\star = \frac{1}{M}\sum_m v_{\phi_m}$*, and a weighted sum with* $\lambda \in \Delta^{M-1}$ *gives* $v_\theta^\star = \sum_m \lambda_m v_{\phi_m}$*. Moreover, because each* $\mathcal{L}_m$ *is a convex quadratic in the velocity in a common isotropic metric,* $\mathcal{A}_L$ *is convex; hence the teacher-mean is* Pareto-optimal in loss*, and no positive-weight scalarizer of* $\mathcal{L}$ *is Pareto-dominated.*

Proof. At a fixed state the integrand $w_j \sum_m \lambda_m \|v - v_{\phi_m}\|_2^2$ is a strictly convex quadratic in v ($w_j > 0$, $\sum_m \lambda_m = 1$); its gradient $2w_j \sum_m \lambda_m (v - v_{\phi_m})$ vanishes uniquely at $v^\star = \sum_m \lambda_m v_{\phi_m}$; $\lambda_m \equiv 1/M$ gives the mean, and state-wise independence makes it global. Convexity of each $\mathcal{L}_m$ in v (shared metric) makes $\mathcal{A}_L$ convex, on which every weighted-sum minimizer is Pareto-optimal by the supporting-hyperplane characterization [1, 13, §4.7.5]. □

Remark 1 (Why the naive convex-hull argument does *not* apply). It is tempting to argue that uniform OPD is dominated because weighted-sum scalarization reaches only the convex hull of the front. Proposition 1 shows this is *false in loss space*: the loss front is convex, so the teacher-mean is genuinely loss-Pareto-optimal and *no* scalarizer is loss-dominated. The real problem is not in loss space at all.

The loss→reward map. The objective we care about is reward, and the reward vector is a *nonlinear, generally non-monotone* functional of the whole matching-loss profile, $R = \Phi(\mathcal{L}, \theta)$, for three concrete reasons: (i) rewards are read off the *integrated* sample x_0, and the ODE/SDE integrator compounds per-step velocity errors nonlinearly, so a lower per-step $\mathcal{L}_m$ need not yield a higher R_m; (ii) reward models are themselves nonlinear (and teachers are imperfect / partially reward-hacked, so *over*-matching a teacher can *lower* its reward); and (iii) a finite, shared LoRA adapter couples the tasks, so capacity spent driving one $\mathcal{L}_m$ toward zero is capacity taken from another. Under such a map, loss optimality does not imply reward optimality.

Proposition 2 (The loss-sum minimizer can be reward-dominated)**.** *There exist achievable OPD students at which the summed-loss minimizer, though loss-Pareto-optimal (Prop. 1), is* not *reward-Pareto-optimal and is reward-dominated by another achievable point.*

Proof by illustrative construction. Consider three achievable students (e.g. three LoRA solutions) with matching-loss vectors $\mathcal{L}$ and reward vectors R:

point	$\mathcal{L} = (\mathcal{L}_1, \mathcal{L}_2)$	$\sum_m \mathcal{L}_m$	$\max_m \mathcal{L}_m$	$R = (R_1, R_2)$
A (teacher-1 specialist)	(1, 8)	9	8	(7, 2)
B (teacher-mean, $\arg\min \sum \mathcal{L}$)	(2, 5)	7	5	(5, 5)
D (min-max, $\arg\min \max \mathcal{L}$)	(4, 4)	8	4	(8, 8)

B uniquely minimizes $\sum_m \mathcal{L}_m$ ($7 < 8 < 9$) and is loss-Pareto-optimal (no point is component-wise $\leq B$: A has $\mathcal{L}_1<$ but $\mathcal{L}_2>$, D is larger in both). Yet B's reward $(5, 5)$ is strictly dominated by D's $(8, 8)$. The transition $B \to D$ *worsens* $\mathcal{L}_1$ ($2 \to 4$) yet *raises* R_1 ($5 \to 8$): reward is a coupled, non-monotone functional (over-serving teacher 2 at B wastes shared capacity and injects its artifacts through ODE integration), so more faithful proxy matching is not more reward. Hence the loss-sum minimizer is reward-dominated, and — critically — the *min-max* student D, which reallocates matching effort toward the worst-served teacher, is the achievable point that lands higher on the reward frontier. □

The next toy shows the same phenomenon on a continuum and explains *why* min-max, not the sum, recovers the balanced high-reward point. We phrase it directly in *reward-gap* space $g_m := R_m^\star - R_m \geq 0$ (gaps to be minimized), whose achievable region is generically *non-convex* because $g = \Phi'(\mathcal{L})$ is nonlinear — this is *not* the convex loss front of Prop. 1.

Proposition 3 (Reward-gap toy: uniform collapses to one teacher; min-max balances)**.** *Let M=2 and let a scalar student control the reward gaps along a* concave *front $g_1(\theta) = \theta$, $g_2(\theta) = 1 - \theta^2$, $\theta \in [0, 1]$, so the achievable gap region is non-convex. Then (a) every weighted sum $\lambda g_1 + (1 - \lambda) g_2$*

is concave in θ and minimized only at an endpoint $\theta^\star \in \{0,1\}$ — the student collapses to a single teacher, and $\lambda{=}\frac{1}{2}$ still returns a corner; (b) the balanced Pareto point $g_1{=}g_2$ is at $\theta^\star = \frac{\sqrt{5}-1}{2} \approx 0.618$, Pareto-optimal yet the minimizer of no weighted sum; (c) the min-max objective $\max\{g_1, g_2\}$ is uniquely minimized at this balanced point.

Proof. (a) $\frac{d^2}{d\theta^2}(\lambda g_1 + (1-\lambda) g_2) = -2(1-\lambda) \le 0$, so the sum is concave and minimized at an endpoint; at $\lambda{=}\frac{1}{2}$ the corners tie at $\frac{1}{2}$ while $\theta{=}\frac{1}{2}$ gives 0.625, so a corner wins. (b) $\theta = 1-\theta^2 \Rightarrow \theta = \frac{\sqrt{5}-1}{2}$; monotonicity of g_1 (up) and g_2 (down) gives Pareto-optimality, and by (a) no weighted sum selects an interior point. (c) $\max\{\theta, 1-\theta^2\}$ is the upper envelope of an increasing and a decreasing branch, quasiconvex with unique minimum at the crossing. □

Remark 2 (Reading of the toys). Proposition 3 is an illustration of the general multi-objective fact *in reward-gap space*; it is deliberately *not* derived from the OPD velocity-matching losses (which are convex and yield a convex front where the teacher-mean is fine, Prop. 1). The non-convexity that makes min-max strictly better is a property of the loss→reward map, which we assert (from the ODE / reward-model / shared-capacity mechanism) and support empirically (§4.6, and the experiments), not a property we prove for the trained SD3.5 student a priori.

4.4 CA-OPD: adaptive Tchebycheff scalarization

We replace the linear sum by a *weighted Tchebycheff* (min-max) scalarization over per-teacher reference-normalized losses $r_m > 0$:

$$\mathcal{L}_\infty(\theta) = \max_{m \in [M]} \frac{\mathcal{L}_m(\theta)}{r_m}. \tag{8}$$

Theorem 1 (Tchebycheff scalarization reaches the full Pareto set; [13, Thm. 3.4.2–3.4.5])**.** *With a utopia reference below the front, for fixed positive weights every minimizer of the weighted Tchebycheff problem is weakly Pareto-optimal, and every properly Pareto-optimal point is the unique minimizer for some positive weight — including points in* non-convex *regions of the front. Ranging over weights, weighted Tchebycheff scalarization characterizes the complete Pareto set, in contrast to the convex-hull restriction of weighted-sum scalarization.*

Equation (8) is nonsmooth; we use the smooth (log-sum-exp) surrogate with temperature $\tau > 0$, and — key for our implementation — give it an exact variational (dual) form.

Proposition 4 (Entropy-regularized min-max and the envelope theorem)**.** *Define the smooth Chebyshev surrogate*

$$\mathcal{F}_\tau(\theta) = \tau \log \sum_{m=1}^{M} \exp\Big(\frac{\mathcal{L}_m(\theta)}{\tau\, r_m}\Big) \;=\; \max_{w \in \Delta^{M-1}} \Big[\sum_{m=1}^{M} w_m \frac{\mathcal{L}_m(\theta)}{r_m} \;+\; \tau\, \mathcal{H}(w) \Big], \tag{9}$$

where $\mathcal{H}(w) = -\sum_m w_m \log w_m$ is the Shannon entropy. The inner maximum is attained uniquely at $w_m^\star = \mathrm{softmax}\big(\mathcal{L}_m/(\tau r_m)\big)$. By the envelope (Danskin) theorem,

$$\nabla_\theta \mathcal{F}_\tau(\theta) = \sum_{m=1}^{M} w_m^\star \, \nabla_\theta \frac{\mathcal{L}_m(\theta)}{r_m}, \tag{10}$$

i.e. the gradient obtained with the weights detached *(treated as constants) equals the* exact *gradient of $\mathcal{F}_\tau$ — no term through $w^\star$ is missed.*

Proof. The dual form of log-sum-exp is the Gibbs variational principle: for $a_m = \mathcal{L}_m/(\tau r_m)$, $\max_{w\in\Delta}[\sum_m w_m(\tau a_m) + \tau\mathcal{H}(w)] = \tau \log \sum_m e^{a_m}$, attained at $w^\star = \mathrm{softmax}(a)$. Since $w^\star$ is the argmax, its first-order contribution vanishes, so Danskin's theorem gives $\nabla_\theta \mathcal{F}_\tau = \sum_m w_m^\star \nabla_\theta(\mathcal{L}_m/r_m)$. □

Proposition 4 makes our detached-weight update *principled rather than approximate*: minimizing $\mathcal{F}_\tau$ is exactly softmax-weighted velocity matching that up-weights the worst-matched (largest $\mathcal{L}_m/r_m$) teacher.

Proposition 5 (Temperature limits)**.** *As $\tau \to 0^+$: $\mathcal{F}_\tau \to \max_m \mathcal{L}_m/r_m$ and $w^\star \to$ one-hot on* $\arg\max$ *— pure min-max (Tchebycheff). As $\tau \to \infty$: $w_m^\star \to 1/M$ and $\nabla\mathcal{F}_\tau \to \frac{1}{M}\sum_m \nabla(\mathcal{L}_m/r_m)$ — the* uniform *DiffusionOPD gradient (mopd), up to the additive constant $\tau \log M$ that does not affect the descent direction.*

Proof. $\tau \log \sum_m e^{a_m/\tau} \to \max_m a_m$ and softmax concentrates on the argmax as $\tau \to 0^+$; as $\tau \to \infty$ the logits $a_m/\tau \to 0$ coincide so $w_m^\star \to 1/M$. □

EMA-softmax adaptive weighting (the deployed CA-OPD). Because per-batch losses are noisy, we weight not by the instantaneous $\mathcal{L}_m$ but by an EMA $\bar{\ell}_m^{(t)} = \beta\bar{\ell}_m^{(t-1)} + (1-\beta)\mathcal{L}_m(\theta^{(t)})$ (β=0.9), and set detached weights

$$w_m^{(t)} = \frac{\exp(\bar{\ell}_m^{(t)}/(\tau r_m))}{\sum_k \exp(\bar{\ell}_k^{(t)}/(\tau r_k))}, \qquad \mathcal{L}_{\text{CA-OPD}}^{(t)}(\theta) = \sum_{m=1}^{M} \mathrm{sg}\big[w_m^{(t)}\big]\, \mathcal{L}_m(\theta). \tag{11}$$

By Proposition 4 this is (variance-reduced, one-step-lagged) descent on $\mathcal{F}_\tau$. The temperature (τ=0.5 in our runs) tunes how aggressively the worst-matched teacher is prioritized; by Proposition 5 mopd is the $\tau \to \infty$ special case, so mopd is contained in CA-OPD's search space. Two honest caveats: (i) the smooth surrogate reaches only an $O(\tau \log M)$ neighborhood of the true Tchebycheff optimum (since $0 \le \mathcal{F}_\tau - \max_m \mathcal{L}_m/r_m \le \tau \log M$); and (ii) the EMA introduces a lag, so (11) equals the exact envelope gradient only in the small-lag limit $\bar{\ell}_m \approx \mathcal{L}_m$.

4.5 Why min-max scalarization is meaningful *in OPD*: output-space commensurability

Generic multi-task learning min-max / Tchebycheff scalarization is fragile because it compares losses of incommensurable units and scales: a cross-entropy and a box-regression loss have no common metric, so $\max_m \mathcal{L}_m$ (or $\mathcal{L}_m/r_m$ with ad hoc r_m) is dominated by arbitrary scaling. OPD removes exactly this pathology, at the level of the model *output*.

Proposition 6 (Shared output metric across teachers)**.** *All M per-teacher losses are the* same functional *— the squared velocity error $\|v_\theta - v_{\phi_m}\|_2^2$ evaluated at the* same *on-policy states with the* same *per-step weights w_j and covariance $\bar{\sigma}_j^2$ — differing only in the target v_{ϕ_m}. Hence the $\mathcal{L}_m$ are measured in one common isotropic metric on velocity-output space and are directly comparable in scale and units: the residual fields $\delta_m := v_\theta - v_{\phi_m}$ live in a single inner-product space, so $\mathcal{L}_m$, the ratios $\mathcal{L}_m/r_m$, and the max in (8) are all well-defined without any per-task rescaling.*

Proof. Immediate from (7): only v_{ϕ_m} carries the teacher index; the metric $w_j \|\cdot\|_2^2$ and the state law p_S (Assumption 1) are shared, so the M losses are values of one functional at M targets. □

This output-space commensurability — not any claim about parameter-space geometry — is what makes min-max / Tchebycheff scalarization meaningful here.

Remark 3 (We do *not* claim meaningful parameter-space conflict angles). One might hope the shared Jacobian $J_\theta = \nabla_\theta v_\theta$ (common to all $\nabla_\theta \mathcal{L}_m = \mathbb{E}[\sum_j 2w_j J_\theta^\top \delta_m]$) makes parameter-space gradient angles $\langle \nabla\mathcal{L}_m, \nabla\mathcal{L}_{m'} \rangle$ directly interpretable. We resist this claim: under LoRA the effective Jacobian is *low-rank and anisotropic*, so $J_\theta^\top J_\theta$ badly distorts angles, and parameter-space conflict is not a clean invariant. Our commensurability argument lives entirely in *output* (velocity) space, where the metric is isotropic and teacher-independent. Conflict-averse gradient methods (MGDA [2, 17], PCGrad [22], CAGrad [11]) are therefore *related* approaches that OPD's commensurability would also make well-posed in output space; but the method we ship is the softmax/Tchebycheff scalarization of §4.4, not a parameter-space gradient surgery.

4.6 Convergence, weighting risk, and empirical framing

Convergence: proven vs. heuristic. For *fixed* scalarizations, Pareto-stationarity is standard: MGDA converges to a Pareto-stationary point [2, 17]; CAGrad converges to a stationary point of the average loss while reducing the worst-case local objective [11]; weighted Tchebycheff minimizers are weakly Pareto-optimal and cover the full front (Theorem 1). For any fixed weight $w^\star \in \Delta^{M-1}$, a stationary point of $\sum_m w_m^\star \mathcal{L}_m$ satisfies $\sum_m w_m^\star \nabla\mathcal{L}_m = 0$, i.e. is Pareto-stationary. Our EMA-softmax scheme (11) is *not* covered by these theorems as stated: the weights are state-dependent and evolve, so CA-OPD is a nonstationary, adaptively scalarized descent without a proven rate. We give only a *conditional* guarantee: *if* the EMA weights converge, $w^{(t)} \to w^\star$ (as observed once per-teacher losses equilibrate), then in the limit CA-OPD is SGD on the fixed weighted objective $\sum_m w_m^\star \mathcal{L}_m$, whose stationary points are Pareto-stationary, and the softmax fixed-point balances the active normalized losses (approximating Theorem 1). We do *not* prove $w^{(t)}$ converges nor give a non-asymptotic rate; a two-timescale stochastic-approximation analysis, or replacing the EMA-softmax by an inner MGDA/CAGrad solve with its existing guarantee, is left to future work. We position CA-OPD as a cheap, principled surrogate for adaptive Tchebycheff scalarization, not a new convergence theorem.

Weighting risk and a regret-based refinement. Min-max on *absolute* losses has a known failure mode: it can chase a teacher whose loss is *irreducible* (intrinsically hard task, or a reward-hacked teacher whose velocity field the student should not fully imitate), perpetually starving the others. The reference r_m partially guards against this, but a cleaner fix is *regret*-based weighting: set r_m or the weighting statistic to the excess over the teacher's own attainable floor,

$$\text{regret}_m(\theta) = \mathcal{L}_m(\theta) - \text{floor}_m, \qquad \text{floor}_m := \min_\theta \mathcal{L}_m(\theta) \text{ (teacher's single-task attainable loss)}, \tag{12}$$

so an intrinsically hard teacher (large floor_m) is not permanently up-weighted, and CA-OPD focuses on *reducible* disagreement. We flag this as the recommended refinement; our reported runs use fixed r_m.

Empirical framing (cross-reference to experiments). Our observed pattern — aesthetics reached *faster* early (the lagging axis), OCR slightly down — is *consistent with* the min-max prediction of reallocating effort toward the worst-served reward axis (Prop. 2, 3); it is *not* a proof of the reward-non-convexity that motivates CA-OPD, and at convergence CA-OPD does *not* exceed mopd on any axis. We disclose the limitations that bound this evidence: results are single-seed at n=256 prompts; the OCR decrease is within run-to-run noise; and we report per-axis metrics only, because a single min-max-normalized average is normalization-dependent and partially circular with a min-max-motivated method. Stronger claims require multi-seed runs and per-axis significance.

4.7 Remark: symmetrizing the OPD divergence is vacuous

Remark 4 (Equal-covariance Gaussians collapse the divergence family to one L_2). Because p_S, p_T share covariance $\bar{\sigma}_j^2 I$ at the same state, forward and reverse KL coincide, $\mathrm{KL}(p_S \| p_T) = \mathrm{KL}(p_T \| p_S) = \|\mu_S - \mu_T\|_2^2/(2\bar{\sigma}_j^2)$, so any convex combination (Jeffreys), the Jensen–Shannon divergence, the squared Hellinger, and the 2-Wasserstein distance are all strictly increasing functions of the single scalar $\|\mu_S - \mu_T\|_2^2 = c_j^2 \|v_\theta - v_\phi\|_2^2$. Every such divergence therefore induces the same minimizer and descent direction up to a positive monotone reweighting: "symmetrizing the KL" is vacuous for OPD. The design freedom lives in *which* targets to match and *how to weight teachers* (this paper), not in the divergence form — a rigor signal that the equal-covariance structure genuinely constrains the space of meaningful modifications. (Scope: this uses both the shared covariance and the shared on-policy state; it would not hold for teacher-rollout states or unequal/learned covariances.)

5 Experiments

We (i) reproduce DiffusionOPD on SD3.5-Medium and confirm it matches the original (§5.2), then (ii) compare CA-OPD against uniform DiffusionOPD at matched training steps, showing a convergence-efficiency benefit but comparable final per-axis quality (§5.3). The four-idea ablation is in §6.

5.1 Setup

We follow DiffusionOPD [10] and DiffusionNFT [23]. The base model is SD3.5-Medium [3] at 512×512; the student is a LoRA adapter (r=32, α=64) trained by on-policy distillation and evaluated with a 40-step first-order ODE sampler. We use three task teachers—GenEval, OCR, and Aesthetics—trained once per task, and distill them into one student. Rewards are the base method's suite: rule-based GenEval [4] and OCR, and model-based PickScore [8], ClipScore [6], HPSv2.1 [20], and Aesthetics [15], with GenEval/OCR on their FlowGRPO splits and model-based rewards on DrawBench [14]. Unless noted, evaluation uses n=256 prompts at student guidance scale 4.5. We report *per-axis* metrics only. A single min-max–normalized average (as in the base method's **Average** column) is normalization-dependent—its value shifts with the choice of reference floor/ceiling, so it is not comparable across papers—and is additionally circular with a min-max method (§4.6); we therefore compare per axis, on which our reproduction matches DiffusionOPD (Table 1).

5.2 Faithful reproduction of DiffusionOPD

Table 1 reports our reproduced base model, single-task teachers, and distilled student (our uniform `mopd` checkpoint at step 721) alongside the published numbers. All rows agree within evaluation noise: the base SD3.5-M (with/without CFG), each teacher's in-domain peak (GenEval 0.97, OCR 0.94, Aes 6.22/HPS 0.347), and the distilled student across all six axes. Our student additionally cross-checks against the released official student (≈ identical), giving a triple-validated reproduction.

5.3 CA-OPD vs. uniform DiffusionOPD

We compare CA-OPD (τ=0.5, β=0.9) against uniform `mopd` at *matched* training steps. The two arms share teachers, data, rollout, and optimizer, differing only in the per-teacher weighting, so any difference is attributable to the reweighting. Table 2 reports both an early checkpoint (@541) and convergence (@721/@781); Figure 1 plots the training-reward proxy and the evaluation trajectory.

Table 1: **Faithful reproduction of DiffusionOPD** on SD3.5-Medium. "Paper" values are from [10] (Table 1); "Ours" are our reproduction (n=256/512). "`mopd`@721" is our reproduced uniform DiffusionOPD student; it matches the paper and the released official student. Dashes: metric not separately re-measured for that row.

Model	Src	GenEval	OCR	Pick	Clip	HPS	Aes
SD3.5-M (w/o CFG)	Paper	0.24	0.12	20.51	0.237	0.204	5.13
	Ours	0.32	0.11	20.6	0.238	0.201	5.18
+ CFG	Paper	0.63	0.59	22.34	0.285	0.279	5.36
	Ours	0.59	0.56	22.4	0.286	0.283	5.42
GenEval Teacher	Paper	0.96	0.40	22.04	0.274	0.248	5.24
	Ours	0.97	0.42	22.0	–	–	–
OCR Teacher	Paper	0.65	0.93	22.27	0.290	0.272	5.26
	Ours	–	0.94	22.3	–	–	–
Aes Teacher	Paper	0.49	0.59	24.02	0.295	0.346	6.22
	Ours	–	–	24.0	0.296	0.347	6.22
DiffusionOPD (Ours=`mopd`@721)	Paper	0.96	0.94	23.99	0.297	0.342	6.15
	Ours	0.958	0.939	23.98	0.298	0.342	6.172

Convergence efficiency (the win). CA-OPD's target is the lagging axis. At the *early* step 541, CA-OPD already reaches an aesthetic score of 6.172—the value uniform `mopd` only attains at its own best checkpoint 721—i.e. roughly 180 steps earlier (Table 2, Fig. 1). The same early checkpoint also leads on GenEval (0.955 vs. `mopd` 0.947), with a small OCR trade (0.927 vs. 0.939). This is the concrete, honest benefit: faster convergence on the hardest-to-match reward axis, exactly the min-max reallocation predicted in §4 (lagging axis up, saturated OCR axis slightly down).

Final quality (no win, stated plainly). By convergence the picture reverses per axis: uniform `mopd` keeps improving while CA-OPD plateaus early. At step 721/781, CA-OPD is within single-seed n=256 noise of `mopd` on every axis and *exceeds it on none*—GenEval, Pick, Clip, and HPS are tied, while OCR (0.928 vs. 0.939) and Aesthetics (6.157/6.145 vs. 6.172) are marginally lower. CA-OPD's aesthetic peak in fact drifts down after its early maximum ($6.172 \to 6.157 \to 6.145$) as min-max continuously re-chases the momentary worst axis. **CA-OPD does not improve final per-axis quality on this benchmark.** The temperature τ is thus best read as a speed-vs-final-quality knob: lower τ accelerates the lagging axis but plateaus sooner; $\tau \to \infty$ recovers `mopd`'s slower-but-higher trajectory.

5.4 Comparison with multi-task RL baselines

Table 3 places the OPD-distilled students against *direct* multi-task reinforcement learning—the DiffusionOPD baselines Multi-Task NFT and Multi-Task GRPO-Guard, each jointly trained on the same three reward tasks by alternating datasets. Consistent with the original paper, the OPD-distilled students dominate the RL baselines on *every* axis: uniform-mopd and CA-OPD are higher than both Multi-Task GRPO-Guard (@960) and Multi-Task NFT on GenEval, OCR, PickScore, ClipScore, HPS, and Aesthetics alike (e.g. Aes 6.15–6.17 vs. 5.64/4.91; PickScore 24.0 vs. 22.9/21.3). Notably, our Multi-Task NFT run *collapses* after step ∼570 (every metric falls to near-zero by step 690), consistent with DiffusionNFT's documented susceptibility to reward-hacking [23]; we therefore report its best (pre-collapse) checkpoint, and this instability is itself a motivation for OPD-style

Table 2: **CA-OPD vs. uniform DiffusionOPD at matched steps** (n=256, gs 4.5; per-axis metrics only). CA-OPD leads early (@541, driven by the aesthetics axis, **bold**) but does not exceed `mopd` on any axis at convergence (@721/@781); per-axis differences at convergence are within single-seed noise.

Model	GenEval	OCR	Pick	Clip	HPS	Aes
Early (step 541)						
`mopd`@541	0.947	0.939	23.92	0.297	0.340	6.120
CA-OPD@541	0.955	0.927	23.95	0.297	0.341	**6.172**
Converged (step 721 / 781)						
`mopd`@721	0.958	0.939	23.98	0.298	0.342	6.172
CA-OPD@721	0.959	0.928	23.96	0.296	0.342	6.157
CA-OPD@781	0.953	0.947	24.00	0.296	0.343	6.145

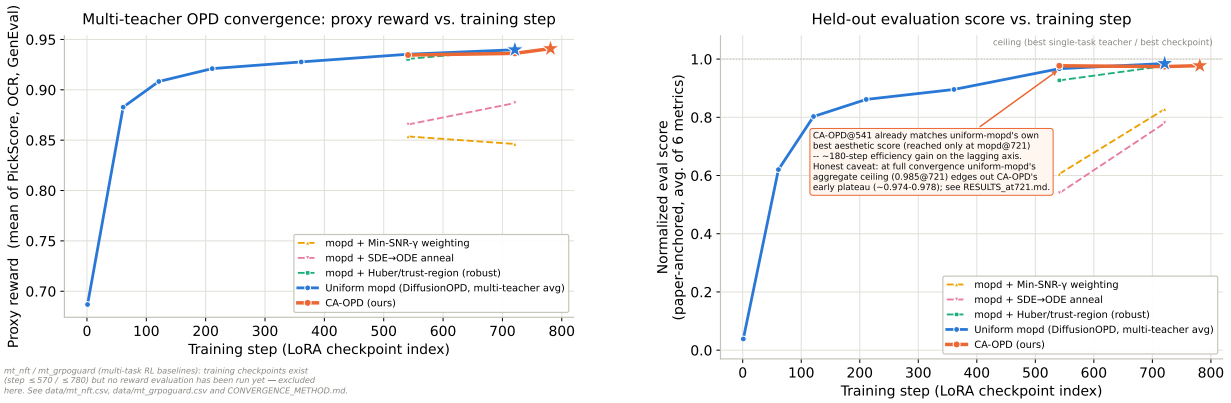


Figure 1: **Convergence of CA-OPD vs. uniform DiffusionOPD.** Left: training-reward proxy vs. steps. Right: a normalization-dependent aggregate evaluation score vs. steps, shown for *trend only* (we compare per axis in the tables). CA-OPD (min-max reweighting) rises faster early and reaches the lagging aesthetic axis ∼180 steps sooner, but plateaus while uniform `mopd` keeps improving to a slightly higher final per-axis quality.

distillation over direct multi-task RL. Multi-Task GRPO-Guard trains stably and is still slowly improving (a conservative lower bound). For reference, the original paper's own converged *Average* scores—on *its* normalization, which we do not reproduce for our runs—are 0.763 (GRPO-Guard) and 0.715 (NFT) vs. DiffusionOPD 0.929, telling the same qualitative story. The ordering **OPD distillation ≫ direct multi-task RL** is unambiguous per axis, which is the premise CA-OPD builds on.

Qualitative comparison. Figure 2 shows cherry-picked samples for shared prompts. CA-OPD's early checkpoints render more polished, aesthetically richer images than same-step `mopd`, consistent with the accelerated aesthetics axis; differences at convergence are subtle, matching the near-tied per-axis numbers.

Table 3: **OPD distillation vs. direct multi-task RL** (n=256, gs 4.5; per-axis metrics only). Multi-task RL baselines are at their latest available checkpoints and still converging (lower bounds). The OPD-distilled students exceed both RL baselines on every axis.

Method	GenEval	OCR	Pick	Clip	HPS	Aes
Multi-Task NFT (@570)	0.639	0.854	21.34	0.280	0.240	4.907
Multi-Task GRPO-Guard (@960)	0.757	0.849	22.94	0.291	0.308	5.637
uniform DiffusionOPD (Ours, @721)	**0.958**	0.939	23.98	**0.298**	0.342	**6.172**
CA-OPD (@781)	0.953	**0.947**	**24.00**	0.296	**0.343**	6.145

6 Ablation: mapping the OPD-improvement design space

Beyond CA-OPD (min-max reweighting), we tested three other principled modifications to multi-teacher OPD. All four share the reproduction setup of §5.1 and differ from uniform `mopd` only in the studied component. Table 4 reports all arms at convergence (step 721); the outcome maps which OPD modifications help and which hurt, and *why*.

(1) Min-max reweighting (CA-OPD). The method of §3. *Helps early* (Table 2: faster on the lagging aesthetics axis) but at convergence is within noise of `mopd` on every axis and higher on none (Aes 6.157 vs. 6.172, OCR 0.928 vs. 0.939). Mechanism: min-max reallocates capacity toward the worst-served reward axis; on a near-saturated benchmark this trades a small final quality margin for convergence speed.

(2) Huber / trust-region residual (`robust`). Replace the squared velocity residual by a Huber loss [7], to be robust to teachers that reward-hack. *Neutral*: within noise of `mopd` across axes, and notably the *best* OCR at convergence (0.951). Mechanism: with well-behaved teachers there are few large residual outliers for the Huber threshold to clip, so it neither helps nor hurts overall—a useful confirmation that OPD's plain L_2 is already well-behaved here.

(3) Min-SNR per-step weighting (`snr`). Apply Min-SNR-γ timestep weighting [5] to the per-step OPD loss. *Hurts badly* (GenEval collapses to 0.785, Aesthetics to 5.851). Mechanism: Min-SNR down-weights high-noise steps, but in OPD those early, high-noise steps set the sample's global structure and compositional layout; suppressing them destroys GenEval and structure. This shows OPD's uniform per-step weighting is not an oversight but a good default.

(4) SDE→ODE noise annealing (`anneal`). Start distillation in the stochastic SDE regime and anneal to the deterministic ODE. *Hurts most* (HPS 0.298 and Aesthetics 5.623 degrade sharply). Mechanism: the early high-noise SDE phase injects variance that degrades the student before annealing can help, directly confirming the base paper's finding that the ODE objective dominates SDE.

Takeaway. The design space splits cleanly: reweighting *across teachers* (min-max, Huber) stays within noise of the strong uniform baseline, whereas reweighting *across denoising steps* (Min-SNR) or perturbing the *sampler regime* (SDE annealing) is actively harmful. This localizes where OPD improvements can plausibly come from—teacher combination, not per-step or sampler manipulation—and shows CA-OPD's early-convergence benefit is not a trivial artifact of any reweighting.

Table 4: **Four-idea ablation at convergence** (step 721, n=256, gs 4.5; per-axis metrics only). Min-max (CA-OPD) and Huber (`robust`) are within noise of `mopd` across axes; Min-SNR and annealing clearly hurt (GenEval/Aesthetics and HPS/Aesthetics respectively). No idea exceeds uniform `mopd` on the final per-axis quality.

Model@721	GenEval	OCR	Pick	Clip	HPS	Aes
`mopd` (uniform)	0.958	0.939	23.98	0.298	0.342	**6.172**
CA-OPD (min-max)	0.959	0.928	23.96	0.296	0.342	6.157
`robust` (Huber)	0.948	**0.951**	23.97	0.298	0.342	6.123
`snr` (Min-SNR)	0.785	0.849	23.52	0.298	0.318	5.851
`anneal` (SDE→ODE)	0.888	0.882	23.14	0.297	0.298	5.623

7 Limitations

We state the boundaries of our claims plainly.

- **Final quality not improved.** On this benchmark CA-OPD does not beat uniform DiffusionOPD on final per-axis quality (within single-seed noise on every axis, higher on none). Its demonstrated benefit is convergence efficiency on the lagging axis and τ-controllability, not final quality.
- **Near-saturated, low-conflict benchmark.** The three teachers (GenEval, OCR, Aesthetics) are already near their individual ceilings and only mildly conflict, leaving little frontier for min-max to relocate along. The theory predicts larger gains for *higher-conflict* teacher sets, which we did not have.
- **Statistical strength.** Results are single-seed at n=256 prompts; the OCR trade (-0.012) is within eval noise, and a confounding GenEval rise muddies the clean "gain-aesthetics/pay-OCR" story. We deliberately report *per-axis* metrics only: a single min-max–normalized average is normalization-dependent (not comparable across papers) and partially circular with a min-max method (§4.6). Multi-seed runs with per-axis error bars are needed for significance.
- **No convergence rate.** CA-OPD's EMA-softmax weights are nonstationary; we prove only a conditional Pareto-stationarity guarantee (if the weights converge), not a rate (§4.6).
- **Absolute-loss weighting risk.** Weighting by absolute EMA-KL can chase an irreducible or reward-hacked teacher. Here the up-weighted axis (aesthetics) was reducible, so the pathology did not bite, but it is a real failure mode.

8 Conclusion

We revisited multi-teacher on-policy distillation through multi-objective optimization. Our main contribution is theoretical: multi-teacher OPD losses are one commensurable functional in a shared isotropic output metric; the uniform (teacher-mean) student is Pareto-optimal *in loss*, so the real difficulty is the nonlinear loss→reward map under which the loss-sum minimizer can be reward-dominated; and an adaptive Tchebycheff min-max reweighting whose detached weights give the *exact* surrogate gradient (envelope theorem) is a principled response, with mopd as its $\tau \to \infty$ limit. Empirically, on a faithful, triple-validated reproduction of DiffusionOPD, our CA-OPD improves

convergence efficiency on the lagging reward axis and exposes τ as a speed-vs-ceiling knob, while an honest four-idea ablation maps the OPD-improvement design space (teacher reweighting is safe; per-step and sampler manipulation hurt). We do not claim a new state of the art: on this near-saturated benchmark CA-OPD does not raise final per-axis quality.

Future work. The theory points to two concrete next steps: (i) **higher-conflict teacher sets**, where min-max frontier selection should yield genuine final-quality gains, not just speed; and (ii) **regret-based weighting** $r_m = \mathcal{L}_m - \text{floor}_m$, weighting the *reducible* gap to a teacher's attainable floor rather than absolute loss, to avoid chasing irreducible teachers. A two-timescale analysis of the EMA-softmax dynamics, or an inner MGDA/CAGrad solve inheriting existing guarantees, would close the convergence-rate gap.

References

[1] Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.

[2] Jean-Antoine Désidéri. Multiple-gradient descent algorithm (mgda) for multiobjective optimization. *Comptes Rendus Mathematique*, 350(5-6):313–318, 2012.

[3] Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, Dustin Podell, Tim Dockhorn, Zion English, and Robin Rombach. Scaling rectified flow transformers for high-resolution image synthesis. In *International Conference on Machine Learning (ICML)*, 2024.

[4] Dhruba Ghosh, Hannaneh Hajishirzi, and Ludwig Schmidt. Geneval: An object-focused framework for evaluating text-to-image alignment. *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[5] Tiankai Hang, Shuyang Gu, Chen Li, Jianmin Bao, Dong Chen, Han Hu, Xin Geng, and Baining Guo. Efficient diffusion training via min-snr weighting strategy. In *International Conference on Computer Vision (ICCV)*, 2023.

[6] Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning. *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021.

[7] Peter J. Huber. Robust estimation of a location parameter. *The Annals of Mathematical Statistics*, 35(1):73–101, 1964.

[8] Yuval Kirstain, Adam Polyak, Uriel Singer, Shahbuland Matiana, Joe Penna, and Omer Levy. Pick-a-pic: An open dataset of user preferences for text-to-image generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[9] Quanhao Li, Junqiu Yu, Kaixun Jiang, Yujie Wei, Zhen Xing, Pandeng Li, Ruihang Chu, Shiwei Zhang, Yu Liu, and Zuxuan Wu. Diffusionopd: A unified perspective of on-policy distillation in diffusion models. *arXiv preprint arXiv:2605.15055*, 2026.

[10] Quanhao Li, Junqiu Yu, Kaixun Jiang, Yujie Wei, Zhen Xing, Pandeng Li, Ruihang Chu, Shiwei Zhang, Yu Liu, and Zuxuan Wu. Diffusionopd: A unified perspective of on-policy distillation in diffusion models. *arXiv preprint arXiv:2605.15055*, 2026.

[11] Bo Liu, Xingchao Liu, Xiaojie Jin, Peter Stone, and Qiang Liu. Conflict-averse gradient descent for multi-task learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[12] Jie Liu, Gongye Liu, Jiajun Liang, Yangguang Li, Jiaheng Liu, Xintao Wang, Pengfei Wan, Di Zhang, and Wanli Ouyang. Flow-grpo: Training flow matching models via online reinforcement learning. *arXiv preprint arXiv:2505.05470*, 2025.

[13] Kaisa Miettinen. *Nonlinear Multiobjective Optimization*. International Series in Operations Research & Management Science. Kluwer Academic Publishers, 1999.

[14] Chitwan Saharia, William Chan, Saurabh Saxena, Lala Li, Jay Whang, Emily Denton, Seyed Kamyar Seyed Ghasemipour, Burcu Karagol Ayan, S. Sara Mahdavi, Rapha Gontijo Lopes, Tim Salimans, Jonathan Ho, David J. Fleet, and Mohammad Norouzi. Photorealistic text-to-image diffusion models with deep language understanding. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

[15] Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, et al. Laion-5b: An open large-scale dataset for training next generation image-text models. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks*, 2022.

[16] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. 2017.

[17] Ozan Sener and Vladlen Koltun. Multi-task learning as multi-objective optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[18] Thinking Machines Lab. On-policy distillation. https://thinkingmachines.ai/blog/on-policy-distillation/, 2026.

[19] Yibin Wang, Yuhang Zang, Hao Li, Cheng Jin, and Jiaqi Wang. Unified reward model for multimodal understanding and generation. *arXiv preprint arXiv:2503.05236*, 2025.

[20] Xiaoshi Wu, Yiming Hao, Keqiang Sun, Yixiong Chen, Feng Zhu, Rui Zhao, and Hongsheng Li. Human preference score v2: A solid benchmark for evaluating human preferences of text-to-image synthesis. *arXiv preprint arXiv:2306.09341*, 2023.

[21] Jiazheng Xu, Xiao Liu, Yuchen Wu, Yuxuan Tong, Qinkai Li, Ming Ding, Jie Tang, and Yuxiao Dong. Imagereward: Learning and evaluating human preferences for text-to-image generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[22] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[23] Kaiwen Zheng, Huayu Chen, Haotian He, Guande Chen, Qinsheng Liu, Jun Zhu, and Jianfei Chen. Diffusionnft: Online diffusion reinforcement with forward process. *arXiv preprint arXiv:2509.16117*, 2025.



Figure 2: **Qualitative comparison** (cherry-picked, shared prompts): uniform DiffusionOPD vs. CA-OPD at matched steps. CA-OPD reaches a more polished aesthetic earlier; convergence-step differences are subtle.